meaning-related capacities of LMs.

Limitations

One limitation of this work is that we are only considering behavioral data which makes it difficult to establish a fully causal link between entity tracking capacities and high performance on our task. Entity tracking is a high-level linguistic behavior and many other capacities are necessary for achieving high accuracy on our task. Therefore, we cannot rule out that differences in some other capacity, such as interpreting sentences compositionally (see Bogin 2022 and Bogin et al. 2022 for evidence that GPT-3 and GPT-3.5 models differ in their compositional generalization behavior), are the main driver for the differences in behavior we see across models.

A possible criticism of our setup is that it requires short-term memory capacities that exceed the memory capacities of most, if not all, humans. That is, if we presented humans with the same input as the model, we would not expect them to be able to keep track of the contents of all 7 boxes due to memory limitations. Therefore we are potentially expecting models to do super-human entity tracking, a setup that has been criticized for model evaluations of other linguistic abilities (Lampinen, 2022). We nevertheless believe that our task is justified given the architecture of the evaluated models. Transformer-based models can look back to any token in the entire input sequence within their context window, so a proper comparison between humans and models would be to present humans with the full description in written form and let them re-read the description after being prompted to state the contents of a box. While we did not formally evaluate whether humans have this ability on a larger population, we personally did not have any trouble tracking the contents of boxes when we had access to the written description.

Relatedly, we designed our task such that the entire description fits within the context window of pretrained language models. However, as we mentioned in the introduction, entity tracking is an important ability for understanding long contexts and given the limited context window, our results do not apply to texts whose length exceeds a model's context window, and likely different model architectures will be necessary to perform proper entity tracking for longer texts.

Further, while we found that GPT-3.5 models as well as finetuned T5 models can track entities in our task with higher accuracy than a strong random baseline, our results also indicate that this behavior is not very stable once several operations act on an entity. Our results should therefore not be taken as justification for using these models for critical applications where high accuracy is needed.

Lastly, we only evaluated English models in this work. Given that we showed that even without high lexical overlap between the training and evaluation examples, models can keep track of entities to some extent, it seems likely that our results also apply to other languages. However, whether this actually the case remains an open question.

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